Rank-five Grothendieck rings through multiplicity twenty

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Abstract

We classify the rank-five fusion rings of multiplicity at most twenty that admit categorifications over $\mathbb{C}$, without assuming pivotality. Exactly seventeen of the 40999 rings admit categorifications, each with a unitary realization; the other 40982 have exact general exclusion certificates. The criteria combine arithmetic, central induction and finite higher-indicator systems. A new inert-prime quadratic parity theorem forces sphericality in two exceptional multiplicity-eighteen cases, which are then excluded by exact central-induction certificates. All applications are automatic from the multiplication tensors, checked against the pinned `fusion-ring-census` repository. Data, certificates and independent verification instructions accompany the paper.

Research notice. AI-generated research · independent verification pending. The results, proofs, code and data have not yet been independently verified by human experts or formalized in Lean. Treat them as provisional; automated checks establish only their stated computational scope.

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1 Introduction and results

A fusion ring is a based ring with nonnegative integral structure constants N_{ij}^k, distinguished unit and involutive duality, satisfying Frobenius reciprocity. Its rank is the number of basis elements, and its multiplicity is $\mu(R) = \max_{i,j,k} N_{ij}^k$. Unit coefficients are included, so

$\mu(R) \geq 1$. By a *Grothendieck ring* we mean one admitting a fusion category over $\mathbb{C}$. The question here is existence for a based ring, not classification of all associators realizing it.

Small-rank enumeration provides candidates rather than categorifications. The pinned census [5] and the multiplicity-one analysis of [9] make this distinction concrete. The unrestricted rank-four classification [3] gives fifteen Grothendieck rings without a multiplicity restriction. The finite rank-four methods of [4] combine formal codegrees, sphericalization and central induction to obtain general, not merely pivotal, exclusions. In rank five the main additional issue is to avoid introducing a new argument for each surviving ring. We address it with parameterized necessary conditions and exact certificates whose hypotheses are read from the multiplication tensor.

Theorem 1.1. *Up to based-ring isomorphism, there are* 40999 *rank-five fusion rings with* $\mu \leq 20$. *Exactly the seventeen rings in Table 2 admit fusion-category categorifications over* $\mathbb{C}$. *All seventeen have unitary realizations. Each of the other* 40982 *rings is excluded from general categorification by the supplied criteria. Consequently general, pivotal, spherical, pseudo-unitary and unitary existence give the same classification of rank-five rings in this range.*

There are 28493 *such rings through multiplicity eighteen. The two exceptional rings* $F^{(5)}_{18,5073}$ *and* $F^{(5)}_{18,5076}$ *are excluded by Corollary 7.2 without a pivotal hypothesis. At exact multiplicities nineteen and twenty there are* 5640 *and* 6866 *rings, respectively; every one is excluded by an exact general certificate.*

The numbering convention is fixed in Section 9; at multiplicity eighteen the two subscripts are also their independent lexicographic indices. Their full tensors, dimensions and certificates are supplied. The two exclusions use the new criterion and exact central-induction certificates.

The new theory has three roles. Theorem 5.2 gives a uniform sphericality criterion with only small integer and mod-two calculations; it eliminates the need for large signed-table searches on its domain. Theorem 7.1 uses the four-element residue field to extend parity rigidity to two genuinely different quadratic tensors. Theorem 6.3 turns the central support of $K(C_4, 4k)$ into a finite obstruction compatible with the same relative-trace framework. The general finite-indicator criterion is Theorem 4.2. The rank-four obstructions used within it or within sphericalization are stated with references, not reproved. The classification is completed by a single exhaustive application and by tensor matches to positive constructions, rather than by separate negative case proofs.

2 Census input and commutativity

The input is the pinned `fusion-ring-census` release [5]. It counts based-ring isomorphism classes, rather than raw multiplication tables; its exact-multiplicity counts through twenty are displayed in Table 1. Our exclusion ledger addresses each rank-five tensor through eighteen. To connect this archived ledger with the repository tables, we canonically flatten every tensor under all 24 permutations of the four nonunit basis elements. The resulting sets agree at each multiplicity $1 \leq \mu \leq 18$: all 28493 repository tensors have an isomorphic archived tensor, and conversely. The multiplicity-nineteen and -twenty certificates address the repository tensors directly, in their recorded repository order. Labels do not generally agree across these releases, so the classification transport uses based-ring isomorphism, never equality of row indices or of labeled-tensor fingerprints. The census identifies the input; the categorical conditions and certificates decide its members.

The following argument supplies the commutativity needed for both the ordinary and signed sphericalization computations. The unsigned assertion also appears in [1, Proposition 9.2].

Lemma 2.1. *Every rank-five fusion ring is commutative. A rank-five signed Frobenius based ring admitting a nonzero, dual-equal dimension character is also commutative.*

Proof. The complex algebra of a (possibly signed) Frobenius based ring is semisimple: the coefficient-of-unit functional gives the positive inner product making the basis orthonormal and multiplication a $*$-representation. A noncommutative algebra of dimension five with a one-dimensional representation must be $\mathbb{C} \oplus M_2(\mathbb{C})$. Its unique one-dimensional character is Galois invariant, hence rational, and its values are algebraic integers, hence integers. For a fusion ring this character is FP; in the signed assertion it is the given dimension character. In either case its formal codegree is an integer $D \geq 5$, the sum of five nonzero integer squares. If f denotes the formal codegree of the two-dimensional representation, the standard identity $\sum_E \dim(E)/f_E = 1$ gives

$$f = \frac{2D}{D-1} = 2 + \frac{2}{D-1}.$$

Formal codegrees are algebraic integers. The displayed rational number is not an integer for $D \geq 5$, a contradiction. □

In the self-dual case, our tensor coordinate convention is the twenty ordered nonunit triples

$$111, 112, 113, 114, 122, 123, 124, 133, 134, 144, 222, 223, 224, 233, 234, 244, 333, 334, 344, 444.$$

The coefficients v_j are listed in this order; multiplication with the unit is fixed by the based-ring axioms. The two tensors resolved by inert-prime parity in Section 9 are recorded in these coordinates. For nonself-dual inputs, the data contain the full multiplication tensors.

μ	Number of rings	μ	Number of rings
1	16	11	1383
2	37	12	1948
3	82	13	2211
4	134	14	2554
5	209	15	2936
6	336	16	4157
7	477	17	4255
8	733	18	5080
9	863	19	5640
10	1082	20	6866

Table 1: Rank-five based-isomorphism classes at exact multiplicity, from the pinned census [5]. The categorification decisions in this paper cover every displayed multiplicity.

3 Categorical inputs and a common certificate framework

Let $N_i = (N_{ij}^k)_{j,k}$ and

$$B = \sum_i N_i N_{i^*}, \qquad p_R(t) = \det(tI - B).$$

For a commutative fusion ring the eigenvalues of B, counted with multiplicity, are its formal codegrees. The same convention will be used for a commutative signed Frobenius quotient. We write $d_i = \mathrm{FPdim}(X_i)$, $D = \sum_i d_i^2$, and use $F : \mathcal{Z}(\mathcal{C}) \to \mathcal{C}$ and I for forgetful and induction whenever $\mathcal{C}$ is spherical.

3.1 Arithmetic and sphericalization

We use the following necessary conditions from [4, Section 2]. Formal codegrees of a complex fusion category are d-numbers. For a monic minimal polynomial $t^r + a_1 t^{r-1} + \cdots + a_r$, this is the integer divisibility condition $a_r^j \mid a_j^r$ for $1 \le j \le r$. Fusion eigenvalues are cyclotomic integers. For a based fusion subring $S \subset R$, the quotient $\mathrm{FPdim}(R)/\mathrm{FPdim}(S)$ is an algebraic integer. The normalized-norm and finite-prime field tests of [4, Lemmas 2.3–2.4] give exact certificates of the required failures. These tests depend on polynomials, divisibility and factorizations over finite fields, not numerical approximations to roots.

Weak integrality permits the pseudo-unitary reduction. More generally, a ring with exactly two distinct simple Frobenius–Perron dimensions has a pseudo-unitary Galois conjugate whenever it has a categorification; we use [4, Proposition 2.1]. In the spherical case a dimension character in the FP Galois orbit gives the same reduction by [4, Lemma 2.2]. Every dimension character is nonzero on simples and equal on dual pairs.

For an arbitrary categorification, pass to the canonical sphericalization and quotient its ring by the relation $\varepsilon = -1$, where $\dim \varepsilon = -1$. The resulting signed coefficients satisfy

$$|T_{ij}^k| \le N_{ij}^k, \qquad T_{ij}^k \equiv N_{ij}^k \pmod 2. \tag{1}$$

The selected lifts may be chosen to have second indicator $+1$ whenever the underlying simple is self-dual. In that normalization the adjoint parity condition becomes

$$T_{jj^*}^i \equiv N_{jj^*}^i \pmod 4 \quad (i = i^*). \tag{2}$$

This is [4, Lemma 7.1 and Corollary 7.2]. We impose no additional sign gauge on the odd coefficients: its compatibility with this normalization would need separate justification. Dual pairs are selected compatibly. The rank-five commutativity argument of Section 2 applies because a spherical dimension character is nonzero on every simple. Thus commutative signed-table enumeration is exhaustive here.

The spherical and signed character tests are exactly those of [4, Sections 2, 6–7]: nonvanishing, equality on dual pairs, formal-codegree norm and field conditions, and the positive central-square moment inequalities, including the central sign object when necessary. A signed table that is a sign gauge of R splits the sphericalization, so the spherical exclusion of R applies. Every other signed character component must receive its own necessary-condition certificate. This is a quantified test over all signed tables and all components, not a conjecture that such tables suffice for associators.

3.2 Low induction and exact moments

In a pseudo-unitary categorification all central dimensions are positive. Since the ring is commutative, the induction of the unit is multiplicity-free. Its simple dimensions are D/f, one for each formal codegree f, and all its twists equal one [12, Theorem 2.13]. If L is the matrix of their restrictions, including the unit column, then

$$L_{0a} = 1, \quad \sum_i L_{ia} d_i = D/f_a, \quad (LL^{\mathsf{T}})_{0i} = B_{0i}, \quad Q = B - LL^{\mathsf{T}} \ge 0 \text{ entrywise}, \quad Q \succeq 0. \tag{3}$$

Column bounds $L_{ia} \le B_{0i}$ and $L_{ia}^2 \le B_{ii}$ make this a finite enumeration. Equal formal-codegree columns may be put in lexicographic order to remove only redundant permutations. The low mass in $I(X_i)$ is $\ell_i = \sum_j (LL^{\mathsf{T}})_{ij} d_j$. The moment bounds of [4, Theorem 3.1 and Proposition 3.2] include

$$(3d_i + 1)\ell_i \le D d_i (d_i + 1) \quad (i = i^*), \qquad 3\ell_i \le D d_i \quad (i \ne i^*). \tag{4}$$

Their low-column implementation is [4, Proposition 3.3]. We retain them as cheap consequences of the common moment framework, not as separate ring-specific methods.

For every positive integer j, the Ng–Schauenburg formula is

$$\sum_{Z\in \operatorname{Irr}\mathcal{Z}(\mathcal{C})} [F(Z) : X_i] \dim(Z)\theta_Z^j = D\nu_j(X_i). \tag{5}$$

Here $\nu_1(X_i) = 0$ for $i \neq 0$; $\nu_2(X_i) = 0$ for a nonself-dual simple and is ± 1 otherwise. If a self-dual simple occurs oddly in some $Y \otimes Y^*$, its second indicator is $+1$ by the adjoint-parity result just cited. Formula (5) is valid for spherical fusion categories [8, Theorem 4.1]; the nonnegative weights used below require the stated pseudo-unitary reduction.

The first tests in this section are inexpensive filters. Their consequences are not claimed to be logically independent or a minimal axiomatization. In particular, we retain an arithmetic rejection when it avoids constructing a much larger, logically stronger moment system.

4 Finite relative-trace certificates

The same construction will be used for integral and quadratic dimensions. In the quadratic case put $K = \mathbb{Q}(\sqrt{q})$, with $q > 1$ squarefree, and let $F = q$ for $q \equiv 1 \pmod 4$, and $F = 4q$ otherwise. In the rational case put $K = \mathbb{Q}$ and $F = 1$. A rational computation is not interpreted as a quadratic field with two embeddings.

4.1 A finite set of joint traces

Let P be a finite set of rational primes containing those dividing F, and fix a height $h \geq 3$. For a root of unity ζ with order supported on P, let

$$\tau_j(\zeta) = \frac{\operatorname{Tr}_{L/K}(\zeta^j)}{[L : K]}, \qquad 1 \leq j \leq h,$$

where L is any common cyclotomic field containing K and ζ. The value is independent of enlarging L.

Lemma 4.1. *All these joint traces are contained in an explicitly computable finite set. It suffices to use orders dividing*

$$B_h = \prod_{p\in P} p^{\max(v_p(F),1)+\lfloor \log_p h\rfloor} \tag{6}$$

and to add the all-zero vector. For $K = \mathbb{Q}$, the trace of a primitive nth root is $\mu(n)/\varphi(n)$. For quadratic K, if χ is the primitive quadratic character of conductor F, then

$$\tau_1(\zeta_n^a) = \frac{\mu(n)}{\varphi(n)} + \begin{cases} \dfrac{\chi(a)\mu(g)\chi(g)\sqrt{F}}{\varphi(n)}, & n = Fg,\ (F, g) = 1, \\ 0, & \text{otherwise}, \end{cases} \tag{7}$$

for $(a, n) = 1$. Higher coordinates are obtained by reducing ζ_n^{aj} to its exact order. The same sign $\chi(a)$ is used in every nonrational coordinate of a joint trace.

Proof. For $F \nmid n$, the intersection $K \cap \mathbb{Q}(\zeta_n)$ is $\mathbb{Q}$, since the conductor of K is F, so the normalized trace is the normalized Ramanujan sum. For $F \mid n$, average over the subgroup on which χ is one. Its indicator is $(1 + \chi)/2$. The untwisted sum is $\mu(n)$, and the induced quadratic Gauss sum is $\chi(a)\mu(g)\chi(g)\sqrt{F}$ when $(F, g) = 1$ and zero otherwise. This gives (7),

including nonsquarefree g. For a power j, set $r = (n, j)$ and apply the formula at order n/r, with primitive exponent aj/r; this proves the simultaneous sign rule.

Suppose $v_p(n) > \max(v_p(F), 1) + \lfloor \log_p h \rfloor$ for some p. For every $j \le h$, its reduced order still has p-valuation strictly greater than both $v_p(F)$ and one. Its Ramanujan sum vanishes, and so does its induced Gauss sum. Hence every coordinate is zero. All other orders divide B_h. Taking both quadratic signs may retain extra tuples when an exponent restriction is redundant; this enlarges the necessary set and cannot create a false exclusion. □

The bound B_h is a bound on *trace types*, not on the exponent of an unknown category. The categorical Cauchy theorem [2, Theorem 3.9], applied to the modular center, allows P to be the prime divisors of $|N_{K/\mathbb{Q}}(D)|$ (of D when $K = \mathbb{Q}$). Thus unbounded possible twist orders cause no omitted trace types.

4.2 Gram columns and indicator bounds

Fix a low matrix L satisfying (3). Each remaining central simple has a restriction vector $v = (0, v_1, \dots, v_{r-1})$ with nonnegative integer entries, and dimension $z_v = \sum_i v_i d_i$. The finite set of permitted vectors contains every actual vector if we require

$$v_i v_j \le Q_{ij}, \qquad Q - vv^{\mathsf{T}} \succeq 0, \qquad z_v \text{ a } d\text{-number}, \qquad D/z_v \text{ an algebraic integer}. \tag{8}$$

Indeed, the residual Gram matrix is a sum of the rank-one matrices of all remaining restriction vectors. The dimension conditions are the modular-center consequences recalled in [4, Proposition 2.1]. Finiteness follows from $v_i \le \lfloor \sqrt{Q_{ii}} \rfloor$. Positive semidefiniteness is checked by principal minors; for a singular Q one also imposes its rational kernel before using a nonsingular principal block. All calculations are rational or integral.

For $j \ge 3$ let

$$H_{i,j} = [X_i^j : 1], \qquad \mathcal{J} = \{3 \le j \le h : K = \mathbb{Q} \text{ or } F \nmid j\}.$$

The indicator is the trace of the order-j rotation on an $H_{i,j}$-dimensional space. Its normalized relative trace to K is rational for $j \in \mathcal{J}$, because $K \cap \mathbb{Q}(\zeta_j) = \mathbb{Q}$. Denote it by $\bar{\nu}_j(X_i)$. If $p(j)$ is the least prime divisor of j, then

$$-\frac{H_{i,j}}{p(j) - 1} \le \bar{\nu}_j(X_i) \le H_{i,j}. \tag{9}$$

To see this, each normalized trace of a rotation eigenvalue is $\mu(n)/\varphi(n)$ for some $n \mid j$. If negative, its denominator is at least $p(j) - 1$. Summing the $H_{i,j}$ terms proves the bounds. No numerical approximation to an indicator is involved.

For every permitted vector v and joint trace type τ introduce a nonnegative real variable $x_{v,\tau}$. It relaxes the integer number of central simples having that data. The necessary equations are

$$\sum_{v,\tau} x_{v,\tau} v_i v_j = Q_{ij}, \tag{10}$$

$$\ell_i + \sum_{v,\tau} x_{v,\tau} v_i z_v \tau_1 = 0, \tag{11}$$

$$\ell_i + \sum_{v,\tau} x_{v,\tau} v_i z_v \tau_2 = D\epsilon_i. \tag{12}$$

Here ϵ_i runs over every second-indicator choice allowed in Section 3. For $j \in \mathcal{J}$, also require

$$\ell_i + \sum_{v,\tau} x_{v,\tau} v_i z_v \tau_j = D u_{i,j}, \qquad u_{i,j} \in \mathbb{Q}, \quad -H_{i,j}/(p(j) - 1) \le u_{i,j} \le H_{i,j}. \tag{13}$$

The rationality condition is a linear equation over $\mathbb{Q}$: writing $D = D_0 + D_1\sqrt{q}$ and the left side as $w_0 + w_1\sqrt{q}$, it is $D_1 w_0 - D_0 w_1 = 0$. The bounds can then be imposed on w_0/D_0, where $D_0 > 0$. For $K = \mathbb{Q}$ the rationality equation is void. Expanding all field coefficients, adding nonnegative slack variables, and clearing denominators gives an integer system

$$Ax = b, \qquad x \geq 0. \tag{14}$$

Zero rationality rows may be retained in the rational implementation.

Theorem 4.2 (Finite-indicator criterion)**.** *Let R be a commutative fusion ring whose FP dimension field is rational or real quadratic. Enumerate all low matrices, all second-indicator choices and all high columns as above. If each low matrix either fails a condition in Section 3, or, for every remaining indicator choice, its system (14) has an integer vector y satisfying*

$$A^{\mathsf{T}} y \geq 0, \qquad b^{\mathsf{T}} y < 0, \tag{15}$$

then R has no pseudo-unitary categorification.

Proof. A hypothetical categorification supplies one of the low matrices and one of the indicator choices. Its remaining restriction vectors satisfy (8); Lemma 4.1 contains all their joint traces. Equations (10)–(13) follow from adjunction and (5), and the indicator bounds were just proved. Consequently it supplies a nonnegative solution x. But $0 \leq x^{\mathsf{T}} A^{\mathsf{T}} y = b^{\mathsf{T}} y < 0$, a contradiction. □

All applications use height six. Retaining higher rational-order bounds contains the third-indicator inequality rather than introducing another criterion. The two-moment inequalities (4) are kept as cheap initial consequences. A floating-point linear program may propose a dual ray; only the cleared integer vector and the exact inequalities (15) are accepted. Failure to find such a ray is not a positive categorification result.

5 Quadratic parity rigidity

This section removes a nonpivotal ambiguity without enumerating signed tables. It is stated for arbitrary self-dual rank; its application to rank five checks only tensor identities and mod-two matrices.

Lemma 5.1 (Dimension-field spectral bound)**.** *Let R be a self-dual fusion ring, and let $\mathcal{C}$ be a hypothetical complex categorification. Select the self-dual lifts in its sphericalization, write their dimensions as $d_0 = 1, d_1, \ldots, d_n$, and set*

$$K = \mathbb{Q}(d_1, \ldots, d_n), \qquad \delta = \sum_i d_i^2.$$

Suppose a formal codegree f of R has minimal polynomial $m_f(t) \equiv t^{\deg m_f} \pmod 2$. Then

$$[K : \mathbb{Q}] \leq \operatorname{ord}_{t=0}(p_R(t) \bmod 2).$$

Proof. The lifts are self-dual: their dimensions are opposite and nonzero, whereas spherical duality preserves dimension. Their signed quotient has commuting real symmetric multiplication matrices T_i and satisfies (1). It is commutative because every element of its basis is self-dual and duality is an anti-automorphism. It is semisimple since its real regular representation is closed under transpose.

Every formal codegree f of R belongs to K, and $\delta/f \in \mathcal{O}_K$. Indeed, inflate the corresponding character to the sphericalization. Its formal codegree is $2f$ and the sphericalization

has dimension 2δ. By [12, Theorem 2.13], a central simple has dimension $2\delta/(2f) = \delta/f$. On forgetting, this is an integral linear combination of the selected dimensions and their negatives. It is an algebraic integer, and is nonzero, so $f \in K$ as well.

Put $e = [K : \mathbb{Q}]$. The Galois orbit of the dimension character has exactly e characters, and contributes the monic integral factor

$$Q_d(t) = \prod_{\sigma : K \hookrightarrow \mathbb{C}} (t - \sigma(\delta)) \tag{16}$$

to the characteristic polynomial of $\sum_i T_i^2$. Its degree is e, with character multiplicity, even when δ generates a proper subfield of K. Integrality follows from Galois invariance and algebraic integrality of the eigenvalues; the complementary Galois-stable factor is integral for the same reason. In a normal overfield, reduce at any prime above two. All conjugates of f reduce to zero by the hypothesis on m_f. Integrality of every conjugate of δ/f implies that all conjugates of δ do too. Thus $Q_d(t) \equiv t^e \pmod 2$. Finally, $T_i \equiv N_i \pmod 2$ gives $t^e \mid \det(tI - \sum_i T_i^2) \bmod 2 = p_R(t) \bmod 2$, as required. □

Theorem 5.2 (Quadratic parity rigidity)**.** *Let R be self-dual of rank $n+1$, with FP dimensions*

$$s_0 = 1, \qquad s_i = A_i + \sqrt{q} \quad (1 \le i \le n),$$

where $q > 1$ is squarefree, $q \equiv 2$ or $3 \pmod 4$, and $A_i \in \mathbb{Z}_{\ge 0}$. Put $M = (N_{jj}^i)_{1 \le j,i \le n}$. Suppose

$$1 + \sum_i A_i^2 = nq, \qquad \operatorname{ord}_{t=0}(p_R(t) \bmod 2) = 2, \qquad \ker_{\mathbb{F}_2} M = \mathbb{F}_2(1, \dots, 1). \tag{17}$$

Assume some $A_j > 0$ has irreducible fusion matrix N_j. Then every complex categorification of R admits a spherical structure and has a pseudo-unitary Galois conjugate.

Proof. Use the notation of Lemma 5.1. The FP formal codegree is

$$D = 1 + \sum_i s_i^2 = 2nq + 2\Big(\sum_i A_i\Big)\sqrt{q}.$$

It generates $\mathbb{Q}(\sqrt{q})$, and its monic minimal polynomial is

$$t^2 - 4nqt + 4n^2q^2 - 4q\Big(\sum_i A_i\Big)^2 \equiv t^2 \pmod 2.$$

The lemma gives $[K : \mathbb{Q}] \le 2$, while its proof gives $D \in K$. Consequently $K = \mathbb{Q}(\sqrt{q})$.

Since $\mathcal{O}_K = \mathbb{Z}[\sqrt{q}]$, write $d_i = a_i + b_i\sqrt{q}$ with $a_i, b_i \in \mathbb{Z}$. At least one b_i is nonzero. Comparing irrational coefficients in the signed square equations gives

$$2a_j b_j = \sum_i T_{jj}^i b_i.$$

Let g be the positive gcd of the nonzero b_i, and put $c_i = b_i/g$. Divide before reducing modulo two: $Mc = 0$ in $\mathbb{F}_2^n$. The integer vector c is primitive, so its mod-two image is nonzero. The kernel hypothesis forces all c_i to be odd, hence all $b_i \neq 0$. This division is necessary; reducing b directly could lose an even common factor.

The characters $(1, d_i)_i$ and $(1, d_i')_i$, where prime denotes quadratic conjugation, are distinct common eigenvectors of real symmetric matrices. They are orthogonal, so

$$1 + \sum_i a_i^2 = q \sum_i b_i^2. \tag{18}$$

The spectral-radius inequality $|T_i| \le N_i$ bounds every conjugate dimension by s_i. Therefore

$$|a_i| + |b_i|\sqrt{q} = \max(|d_i|, |d_i'|) \le A_i + \sqrt{q}.$$

Since $|b_i| \ge 1$, this implies $|a_i| \le A_i$. Thus

$$nq \le q \sum_i b_i^2 = 1 + \sum_i a_i^2 \le 1 + \sum_i A_i^2 = nq.$$

Equality holds throughout: $|b_i| = 1$ and $|a_i| = A_i$.

Choose the embedding for which $|d_j| = s_j$ at an irreducible N_j with $A_j > 0$, and set $v = |d|$ entrywise. From $N_j v \ge |T_j d| = s_j v$, a positive left Perron vector forces equality. Irreducibility then gives $v = s$, with the scalar fixed by $v_0 = s_0 = 1$. Put $\eta_i = \operatorname{sgn}(d_i)$, $\eta_0 = 1$, and $U_{ik}^{\ell} = \eta_i \eta_k \eta_\ell T_{ik}^{\ell}$. The vector s is a positive character for both U and N. Hence

$$0 = \sum_{\ell} (N_{ik}^{\ell} - U_{ik}^{\ell}) s_\ell.$$

Each summand is nonnegative by (1); therefore $U = N$.

Interchanging a selected lift with its opposite according to η_i realizes this integer sign change. The selected and opposite multiplicities in a product are $(N + U)/2$ and $(N - U)/2$. The latter vanish, so the selected lifts form a fusion subcategory of the sphericalization, equivalent to $\mathcal{C}$ under the forgetful functor. The spherical structure restricts to it; compare [12, Section 5.1]. Although one embedding was used in the argument, $U = N$ is an equality of integer tensors and constructs the section for the original category. Applying the chosen embedding to its structure data gives positive dimensions s_i, hence the stated pseudo-unitary Galois conjugate. □

Remark 5.3 (Recognition from the tensor). In rank five the vector (A_i) need not be guessed. The irrational coefficient of $s_j^2 = \sum_i N_{jj}^i s_i$ gives $2A_j = \sum_{i=1}^4 N_{jj}^i$. Recover these four integers from the square rows, set $q = (1 + \sum_i A_i^2)/4$, and verify every dimension-product identity. Squarefreeness, the mod-two characteristic-polynomial multiplicity, the kernel and irreducibility are then exact finite tests. The supplement implements this recognition for every census tensor; no radicand or list of exceptional indices is supplied to the criterion.

6 Cyclic order-four central support

Write $K(C_4, 4k)$ for the near-group ring

$$g^4 = 1, \qquad gX = Xg = X, \qquad X^2 = 1 + g + g^2 + g^3 + 4kX,$$

where $k \ge 1$ is an integer. The following reduction is uniform in k. Thornton's theorem [13, Theorem 1.3] permits a pseudo-unitary Galois conjugate of any complex categorification of a near-group ring; no braidedness or unitarity hypothesis is imposed on that reduction. Set

$$d = 2k + 2\sqrt{k^2 + 1}, \quad d^2 = 4 + 4kd, \quad D = 4 + d^2 = 8 + 4kd, \quad t = D/D' = 1 + kd = d^2/4.$$

The dimension d is irrational because $k^2 < k^2 + 1 < (k + 1)^2$.

6.1 A finite central-support lemma

For a central simple Z put

$$F(Z) = \sum_{r=0}^{3} m_{rZ} g^r + b_Z X, \qquad a_Z = \sum_r m_{rZ}.$$

Call Z *mixed* if $a_Z > 0$, and *pure* otherwise. The induction formula and adjunction give

$$\sum_Z m_{rZ} m_{sZ} = 4\delta_{rs} + 1, \qquad \sum_Z m_{rZ} b_Z = 4k, \qquad \sum_Z b_Z^2 = 8 + 16k^2. \tag{19}$$

Lemma 6.1. *The mixed simples consist of six objects B_{rs}, $0 \le r < s \le 3$, and two objects $E_{r,1}, E_{r,2}$ for every r. Their restrictions are*

$$F(B_{rs}) = g^r + g^s + kX, \qquad F(E_{r,1}) = g^r + u_r X, \quad F(E_{r,2}) = g^r + (k - u_r)X,$$

where $u_0 = 0$, $0 \le u_r \le \lfloor k/2 \rfloor$, and $u_1 = u_3$. For some $\eta \in \{i, -i\}$ their twists are

$$\begin{array}{c|c} E_{0,*}, E_{2,*}, B_{01}, B_{02}, B_{03} & 1 \\ B_{12}, B_{23} & -1 \\ B_{13} & \eta \\ E_{1,*}, E_{3,*} & -\eta. \end{array} \tag{20}$$

The set of r for which $u_r = 0$ is a subgroup of C_4. Its size h belongs to $\{1, 2, 4\}$ and equals the number of central invertibles. Writing $L_r = u_r^2 + (k - u_r)^2$, the total pure mass is

$$\sum_{Z \text{ pure}} b_Z^2 = 8 + 10k^2 - \sum_r L_r. \tag{21}$$

If $h \ge 2$, there is a central invertible z with $F(z) = g^2$, $z^2 = 1$ and $\theta_z = 1$. Splitting pure simples by their monodromy sign with z gives

$$\sum_{Z \in \mathcal{P}_+} b_Z^2 = 4 + 6k^2 - \sum_r L_r, \qquad \sum_{Z \in \mathcal{P}_-} b_Z^2 = 4 + 4k^2, \tag{22}$$

$$\sum_{Z \in \mathcal{P}_-} b_Z^2 \theta_Z^j = 0 \qquad \text{for every odd } j. \tag{23}$$

Proof. The formal codegrees are $D, D', 4, 4, 4$. Thus $I(1) = 1 \oplus A \oplus B_{01} \oplus B_{02} \oplus B_{03}$, with $F(A) = 1 + kX$ and $F(B_{0r}) = 1 + g^r + kX$; this follows from their dimensions $1, t, D/4, D/4, D/4$ and (19). All have twist one.

The pointed associator class is trivial. Indeed, $\mathrm{add}(X)$ is a rank-one module category over $\mathcal{C}_{\mathrm{pt}}$; choosing the isomorphisms $g^r X \simeq X$ gives a two-cochain whose coboundary is that associator. Thus the pointed indicators equal those of Vec_{C_4}: $\nu_j(g^r) = 1$ if $g^{rj} = 1$ and zero otherwise. Positivity in (5) at $j = \mathrm{ord}(g^r)$ forces $m_{rZ} > 0 \Rightarrow \theta_Z^{\mathrm{ord}(g^r)} = 1$. All mixed twists therefore belong to $\{1, -1, i, -i\}$.

Apply the four-point Fourier transform to the traces for $I(g^r)$ at $j = 0, 1, 2, 3$, with $j = 0$ denoting dimension. Comparing coefficients of $1, d$ is valid over $\mathbb{Q}(i)$ because $d \notin \mathbb{Q}(i)$. After summing over r the budgets are

θ	1	-1	i	$-i$
$\sum a_Z^2$	16	8	4	4
$\sum a_Z b_Z$	$8k$	$4k$	$2k$	$2k$.

(24)

Every m_{rZ} is zero or one. For $r = 0$ this follows from $I(1)$. For $r \neq 0$, B_{0r} already contributes one to the diagonal Gram entry five. An additional multiplicity two would consume the remaining four; its overlap with either of the other two nonzero rows would then be even, contrary to the off-diagonal entry one in (19).

A support of size three must avoid the unit row and therefore consist of g, g^2, g^3. Its twist is ± 1 and it contributes nine to its a^2-budget. But the negative budget is eight, while $I(1)$ already contributes fourteen to the positive budget sixteen. A support of size four is

already impossible in the unit row. Each pair of rows consequently overlaps in exactly one pair-support object, and each row has two remaining single-support objects.

Neither B_{12} nor B_{23} fits in the remaining positive budget two, so both have twist -1 and exhaust its budget. The two g^2-supported objects must then have twist one. The object B_{13} exhausts one imaginary budget, leaving all four g- and g^3-supported singles with the opposite imaginary twist. This proves (20). The individual Fourier equations for $I(g)$ and $I(g^3)$ give mass $D/4 = 2 + kd$ in each fourth-root sector; those for $I(g^2)$ give $D/2$ in each sign sector. Comparing dimensions forces every pair coefficient to be k and every single pair to have sum k. Duality identifies the unordered coefficient pairs over g and g^3, so $u_1 = u_3$.

Only a single-support object with coefficient zero has dimension one. Forgetting embeds the group of central invertibles into C_4, because the only invertible summand of $I(1)$ is the unit. This proves the subgroup assertion and (21). If $h \geq 2$, its order-two object lies over g^2 and has twist one by (20). Tensoring with it translates supports by two. All eight singles and exactly B_{02}, B_{13} centralize it; the other four pairs have monodromy -1. Their total mixed squared dimension is

$$16 + 16kd + (\sum_r L_r + 2k^2)d^2.$$

The centralizer has dimension $D^2/2$ by [11, Theorem 3.14], and

$$D^2/2 = 16 + 16kd + (4 + 8k^2)d^2.$$

Subtracting proves (22). On the negative sector tensoring with z preserves b_Z and changes the twist to $-\theta_Z$. There are no fixed points, and pairing the terms proves (23). □

Lemma 6.2 (Balanced Galois obstruction)**.** *A branch with $h = 1$ and $u_1 = u_2 = u_3 = k/2$ requires $k^2/8 \in \mathbb{Z}$.*

Proof. Here A is the unique central simple of dimension t. The signed Galois symmetry of the normalized modular matrix S/D [2, (2.13)], at the embedding sending d to d', sends the unit row to the A row with positive sign. Hence $S_{AZ} = t\,\sigma(\dim Z)$. Its eigenvalue ratios $S_{AZ}/\dim Z$ are t, t^{-1} at $1, A$, one at every other mixed simple, and minus one at every pure simple; use $td' = -d$. Since A is self-dual, subtracting the unit fusion coefficient from Verlinde's formula leaves only the two exceptional columns:

$$N_{AA}^Z - \delta_{Z,1} = \frac{(t^2-1)(\dim Z - \sigma(\dim Z))}{D^2} = \frac{k}{4} b_Z.$$

The last equality uses $(t^2-1)(d-d')/D^2 = k/4$. Every single over a nonunit group element has $b_Z = k/2$. Its fusion coefficient must therefore equal the integer $k^2/8$. □

6.2 The parameterized indicator system

Fix a moment height $H \geq 3$. For each u_1, u_2, u_3 in Lemma 6.1 and $\epsilon = \nu_2(X) \in \{-1, 1\}$, take the known mixed data from (20). Put

$$W_j = \sum_{Z \text{ mixed}} b_Z(a_Z + b_Z d)\theta_Z^j.$$

Apply normalized relative trace to $K = \mathbb{Q}(\sqrt{k^2+1})$, interpreted using its squarefree radicand. We may fix $\eta = i$, since the resulting traces are unchanged by $i \mapsto -i$. Assign a nonnegative weight x_τ to each pure joint trace type of Lemma 4.1; it represents the sum of b_Z^2 for that type. Its total is (21). For $h \geq 2$, instead use two pools with the masses (22). In the negative pool replace a joint tuple by

$$(\tau_j)_{j=1}^H \longmapsto (0 \text{ for odd } j,\ \tau_j \text{ for even } j)_{j=1}^H,$$

which is the average of the paired twists $\theta, -\theta$. The first two equations are

$$\mathrm{Tr}^{\mathrm{norm}}_{L/K}(W_1) + d\sum x_\tau \tau_1 = 0, \qquad \mathrm{Tr}^{\mathrm{norm}}_{L/K}(W_2) + d\sum x_\tau \tau_2 = \epsilon D.$$

For orders $3 \le j \le H$ with $F \nmid j$, impose the rationality and bounds (9) on $(\mathrm{Tr}^{\mathrm{norm}}_{L/K}(W_j) + d\sum x_\tau \tau_j)/D$. The needed rotation dimensions are obtained without tensor powers: if $X^j = \alpha_j \sum_r g^r + \beta_j X$, then

$$(\alpha_1, \beta_1) = (0, 1), \qquad (\alpha_{j+1}, \beta_{j+1}) = (\beta_j, 4\alpha_j + 4k\beta_j), \qquad [X^j : 1] = \alpha_j.$$

The prime support is that of $2(k^2+1)$, since $N_{K/\mathbb{Q}}(D) = 64(k^2+1)$. All these systems are rational and become integer systems after row scaling.

Theorem 6.3 (Cyclic-four central-support criterion)**.** *Fix $k \ge 1$ and a finite moment height $H \ge 3$. Enumerate all the coefficient and indicator branches above. If each branch is rejected by Lemma 6.2 or by an integer separating vector (15) for its central-support system, then $K(C_4, 4k)$ admits no fusion categorification over $\mathbb{C}$.*

Proof. Any categorification has a pseudo-unitary Galois conjugate. Its central support satisfies Lemma 6.1, so belongs to an enumerated branch. Lemma 6.2 rules out the indicated balanced branches. Every other branch supplies nonnegative weights in its indicator system by (5), the paired negative sector and Lemma 4.1. An integer separator contradicts their existence exactly as in Theorem 4.2. □

The software recognizes the entire cyclic-four tensor before applying this criterion; recognizing only the number of invertible objects is not sufficient. At height six it rejects the parameters $k = 2, 3, 4, 5$ encountered in the census beyond the positive case $k = 1$. Their allocations are reported together in Section 9; no separate numerical root-of-unity argument is used for one selected ring.

7 Inert-prime quadratic parity and the final two tensors

The preceding parity theorem uses the integral basis $1, \sqrt{q}$ and requires irrational coefficient one. The following extension treats the case in which 2 is inert in the full ring of integers and the Frobenius–Perron dimensions have irrational coefficient two. Its only extra finite step enumerates characters over the four-element field.

Theorem 7.1 (Inert-prime parity rigidity)**.** *Let R be a self-dual rank-five fusion ring with*

$$s_0 = 1, \qquad s_i = A_i + 2\sqrt{q} \quad (1 \le i \le 4),$$

where $q > 1$ is squarefree, $q \equiv 5 \pmod 8$, $A_i \ge 0$, and $1 + \sum_{i=1}^4 A_i^2 = 16q$. Put $M = (N_{jj}^i)_{1 \le j,i \le 4}$ and $B = \sum_{i=0}^4 N_i^2$. Suppose that

1. $\mathrm{ord}_{t=0}(\det(tI - B) \bmod 2) = 2$;
2. $\ker_{\mathbb{F}_2} M = \mathbb{F}_2(1,1,1,1)$;
3. *for every ring homomorphism $\chi : R \otimes \mathbb{F}_2 \longrightarrow \mathbb{F}_4$ with $\chi(1) = 1$ and $\sum_{i=0}^4 \chi(X_i)^2 = 0$, all $\chi(X_i)$ lie in $\mathbb{F}_2$, and exactly three of $\chi(X_1), \dots, \chi(X_4)$ equal one;*
4. *N_j is irreducible for some $j > 0$ with $A_j > 0$.*

Then every complex fusion-category categorification of R is spherical and admits a pseudo-unitary Galois conjugate.

Proof. Assume $\mathcal{C}$ categorifies R. Choose the self-dual lifts in its canonical sphericalization, write their dimensions $d_0 = 1, d_1, \dots, d_4$, and let T_i be the signed multiplication matrices. By (1), $T_i \equiv N_i \pmod 2$ and $|T_i| \le N_i$ entrywise. Their dimension character has field $K = \mathbb{Q}(d_1, \dots, d_4)$ and $\delta = 1 + \sum_i d_i^2$.

Write $D = 1 + \sum_i s_i^2 = 32q + 4(\sum_i A_i)\sqrt{q}$. Since some $A_j > 0$, D generates $\mathbb{Q}(\sqrt{q})$; its quadratic minimal polynomial reduces to t^2 modulo two. Lemma 5.1 and hypothesis (1) show $K = \mathbb{Q}(\sqrt{q})$. We also need the stronger divisibility already implicit in that lemma: the FP character of R inflates to a character of the sphericalization with formal codegree $2D$. The corresponding simple summand of its induced unit has dimension $(2\delta)/(2D) = \delta/D$ by [12, Theorem 2.13]. Central-simple dimensions are algebraic integers, so $\delta/D \in \mathcal{O}_K$. Since $D \in 2\mathcal{O}_K$, this implies $\delta \in 2\mathcal{O}_K$.

Now $\mathcal{O}_K = \mathbb{Z}[\omega]$, $\omega = (1 + \sqrt{q})/2$. Because $q \equiv 5 \pmod 8$, the reduction of ω satisfies $\bar{\omega}^2 + \bar{\omega} + 1 = 0$; thus $\mathcal{O}_K/2\mathcal{O}_K = \mathbb{F}_4$. The residues $\bar{d}_i$ constitute a character χ of R over $\mathbb{F}_4$ because $T \equiv N \pmod 2$, and they satisfy $\sum_i \bar{d}_i^2 = \bar{\delta} = 0$. By (3) they all lie in $\mathbb{F}_2$, with exactly three nonunit residues equal to one. Writing $d_i = x_i + y_i\omega$ with integers x_i, y_i, this forces y_i even. Consequently $d_i = a_i + b_i\sqrt{q}$ for integers a_i, b_i.

The two Galois-conjugate dimension characters are distinct orthogonal eigenvectors of the real symmetric commuting matrices T_i. Their orthogonality reads

$$1 + \sum_{i=1}^{4} a_i^2 = q \sum_{i=1}^{4} b_i^2. \tag{25}$$

Not all b_i vanish, since K is quadratic. Let g be the positive gcd of the b_i, and put $c_i = b_i/g$. Comparing irrational parts in $d_j^2 = \sum_i T_{jj}^i d_i$ gives $2a_j b_j = \sum_i T_{jj}^i b_i$. Divide by g, then reduce modulo two: $Mc = 0$. The vector c is primitive; by (2), every c_i is odd. In particular each $b_i \neq 0$.

If g were odd, then all four b_i would be odd. Reduction in $\mathcal{O}_K/2$ sends $\sqrt{q} = 2\omega - 1$ to one, so $\bar{d}_i = \bar{a}_i + 1$. Since exactly three nonunit $\bar{d}_i$ equal one, exactly one of the a_i is odd. But (25) modulo four yields $1 + \sum a_i^2 \equiv q \sum b_i^2 \equiv 0 \pmod 4$, and would require exactly three odd a_i. This contradiction makes g even; hence $|b_i| \ge 2$ for every i.

The spectral-radius inequality for $|T_i| \le N_i$ gives $|d_i|, |d_i'| \le s_i$, and therefore $|a_i| + |b_i|\sqrt{q} \le A_i + 2\sqrt{q}$. Thus $|a_i| \le A_i$. Hypothesis $1 + \sum A_i^2 = 16q$, together with (25), implies equality throughout: $|a_i| = A_i$ and $|b_i| = 2$ for every i.

Choose the real embedding for which $|d_j| = s_j$ at the irreducible N_j in (4). The Perron-vector argument in the proof of Theorem 5.2, applied verbatim to $|T_j| \le N_j$, then gives $|d_i| = s_i$ for all i. Changing the selected lift signs accordingly gives a signed tensor U with $|U_{ik}^\ell| \le N_{ik}^\ell$ and the common positive dimension character s. Every coefficient of $N - U$ is nonnegative, and its dimension-weighted sum is zero; hence $U = N$. The selected lifts therefore form a spherical fusion subcategory mapping tensor equivalently to $\mathcal{C}$. Under the chosen embedding this structure has positive simple dimensions, proving the claimed pseudo-unitary Galois conjugate. □

The hypotheses are recognizable from the full tensor, without a census label. The irrational coefficients of s_j^2 give $A_j = \frac{1}{2}\sum_{i=1}^4 N_{jj}^i$; the proposed q is then $(1 + \sum A_i^2)/16$. The supplement checks every product identity, squarefreeness, the characteristic polynomial modulo two, the kernel of M, all 4^4 possible residue tuples, and irreducibility.

Corollary 7.2. *Neither $F^{(5)}_{18,5073}$ nor $F^{(5)}_{18,5076}$ admits a categorification over $\mathbb{C}$, with no pivotal hypothesis.*

Proof. For the two tensors in (26), the vectors (A_1, A_2, A_3, A_4) are respectively $(33, 34, 33, 35)$ and $(33, 33, 34, 35)$, and $q = 285$. In both cases the mod-two zero multiplicity of $\det(tI - B)$

is two, and $\ker_{\mathbb{F}_2} M = \mathbb{F}_2(1,1,1,1)$. With $\bar{\omega}^2 + \bar{\omega} + 1 = 0$, all normalized residue characters are as follows; tuples display their four nonunit entries:

	all residue characters	$\sum_i \chi(X_i)^2 = 0$
5073	$(0,1,0,1), (0,1,1,0), (1,0,1,1), (1,1,0,0)$	$(1,0,1,1)$
5076	$(0,0,1,1), (1,1,0,1), (\bar{\omega}, \bar{\omega}^2, 1, 0), (\bar{\omega}^2, \bar{\omega}, 1, 0)$	$(1,1,0,1)$

Theorem 7.1 applies. The independent exact spherical-character certificates supplied for these tensors then exclude the resulting spherical categories. Both finite checks are recomputed from the tensors in the supplement. □

8 Unitary realizations

(μ, j)	Local index	Unitary realization
$(1,1)$	7	Tambara–Yamagami for C_2^2
$(1,3)$	9	$SU(2)_4$
$(1,4)$	14	$\mathrm{Rep}(D_{14})$
$(1,6)$	12	$\mathrm{Rep}(S_4)$
$(1,7)$	11	$PSU(2)_8$
$(1,10)$	15	$PSU(2)_9$
$(1,11)$	2	Tambara–Yamagami for C_4
$(1,13)$	4	The dimension-twelve model of [9]
$(1,14)$	5	The dimension-twenty-four model of [9]
$(1,16)$	1	$\mathrm{Vec}(C_5)$
$(2,27)$	33	$\mathrm{Rep}(A_5)$
$(2,36)$	5	$PSU(3)_4$
$(2,37)$	1	$\mathrm{Rep}(C_7 \rtimes C_3)$
$(3,40)$	28	C_2-equivariantization of an Izumi $C_3 + 6$ category
$(3,75)$	8	$\mathrm{Rep}(\mathrm{AGL}(1,5))$
$(4,55)$	20	Izumi near-group $C_2^2 + 4$
$(4,123)$	6	Izumi near-group $C_4 + 4$

Table 2: The seventeen positive rings. D_{14} has order fourteen. Each tensor is constructed and matched by a unit-fixing permutation; the supplied permutation is checked.

The pointed, Tambara–Yamagami, finite-group and quantum-group entries use their standard unitary realizations. The two nonstandard multiplicity-one entries are the tensors of [9, Section 3.5], with the unitary constructions in its Sections 4.2 and 4.4. The $PSU(3)_4$ tensor agrees with the modular data in [6]. The near-group realizations for $C_4 + 4$ and $C_2^2 + 4$ are [7, Examples 9.5–9.6]. These references establish existence, not negative classifications used as an oracle.

The software constructs the $SU(2)$ tensors by the truncated Clebsch–Gordan rule, the finite-group tensors by exact character products, and the remaining displayed tensors in the cited conventions. It checks their ring axioms, distinctness and a full tensor isomorphism to the pinned census tensors. The unitary existence is supplied by the constructions, not inferred just from passing the ring axioms.

For completeness, we explain the equivariantization entry, since an order-two outer symmetry by itself would not supply the required action. In [7, Example 13.10], the $C_3 + 6$ Cuntz realization has actual symmetries generated by the pairs $(-1, R_{\pi/4})$ and (id, L), where $L = \mathrm{diag}(1,-1)$ and $R_{\pi/4}$ is planar rotation. Their product is $(-1, U)$ with $U = R_{\pi/4}L$ and $U^2 = I$. Thus it gives an actual order-two automorphism of the realization, fixing the noninvertible endomorphism and the unit and interchanging the two nontrivial invertibles. It

fixes the unit inclusion in the square of that endomorphism. On its six-dimensional self-fusion multiplicity space $\mathbb{C}[C_3] \otimes \mathbb{C}^2$ it is inversion tensored with the reflection U, and has trace zero. The two eigenspaces have dimension three.

The C_2-equivariantization has simples $1, g, V, X, Y$: two equivariant structures over the unit and the noninvertible object, and one object over the orbit of the two other invertibles. The fixed unit inclusion makes X, Y self-dual. Tensoring equivariant structures and using the two three-dimensional eigenspaces gives

$$\begin{aligned} g^2 &= 1, & gV &= V, & gX &= Y, \\ V^2 &= 1 + g + V, & VX &= VY = X + Y, & & \\ X^2 = Y^2 &= 1 + V + 3X + 3Y, & XY &= g + V + 3X + 3Y. & & \end{aligned}$$

The action is unitary, so its equivariantization is unitary. Its tensor is generated from these products and matched exactly, just like the other sixteen models.

9 Automatic application and complete classification through twenty

9.1 A label-free decision procedure

Every exclusion receives its input as the complete integer multiplication tensor. The application order is arithmetic screening, the cyclic-four central-support criterion when recognized, both quadratic parity criteria when their hypotheses hold, and otherwise the positive-indicator sphericalization framework. Pseudo-unitary branches use the low-induction bounds followed by the same finite-indicator systems. Spherical branches exhaust the dimension-character components. Signed branches exhaust the bounded associative tables in the normalization (2) and every eligible character component. Successful branches are rechecked before a certificate is accepted.

None of these functions receives a multiplicity index, historical label, previous status or positive-list complement as a hypothesis. The seventeen model matches are a separate positive channel. An arithmetic failure, field obstruction, missing complete low matrix, exact moment violation or integer separating vector is a mathematical witness. A search timeout, unsupported field or unsuccessful linear program is not. This policy is enforced by the decision ledger.

The top-level allocations in Table 3 are disjoint and follow the actual decision order. Formal-codegree arithmetic comes first because it settles almost all inputs without a central search. The cheap cyclotomic and fusion-subring checks follow. Exact recognition of cyclic-four and quadratic-parity tensors is cheap; their specialized systems are built only on those families. Positive reduction and central induction precede the full signed sphericalization search. These choices balance rejection frequency and cost, without claiming a formal runtime optimality theorem. The low-moment bounds are retained for their cost advantage, despite being consequences of stronger moment conditions.

The last new *criterion family* first used in the checked range is inert-prime parity at multiplicity eighteen, where it resolves two exceptional tensors. No new family or combined certificate type is first needed at multiplicities nineteen or twenty. This is an exact observation about the verified ledger and identifies the earliest multiplicity after which the present range uses no new criterion. It makes no assertion for larger multiplicities.

For the cyclic-four tensors, automatic recognition returns $k = 2, 3, 4, 5$. The height-six criterion checks respectively $6, 6, 14, 14$ support/indicator branches. At $k = 2$ the balanced Galois test removes two and exact separators remove four; at $k = 3, 4, 5$ all branches have

First accepted top-level certificate	First μ	$\mu \le 17$	$\mu \le 18$	$\mu \le 20$
Formal-codegree d-number test	1	23075	28124	40558
Cyclotomicity test	5	5	6	8
Fusion-subring index test	3	12	12	14
Cyclic-four central support	8	3	3	4
Quadratic parity and positive induction	3	12	12	20
Quadratic parity and spherical characters	12	6	6	6
Inert parity and spherical characters	18	0	2	2
Positive reduction and central induction	2	27	29	30
Positive-indicator sphericalization	2	256	282	340
Total general exclusions		23396	28476	40982

Table 3: Disjoint first-accepted certificates in executable order. First use at a multiplicity does not date the underlying theorem. The order favors cheap general tests, then structural recognition before restricted and full central-induction searches.

exact separators. The same routine leaves the positive $k = 1$ model unexcluded. These are recorded applications of Theorem 6.3, not three separate nonexistence arguments. Similarly, the twenty-six top-level quadratic-parity applications are detected from the square rows and verified modulo two; no fixed list of radicands enters the code. The two inert-prime applications at eighteen additionally enumerate all four-element-field residue characters and verify the central divisibility filter before invoking their spherical certificates.

Integral inputs use the rational specialization of Theorem 4.2. No published classification of integral rings is invoked to exclude them. No one-ring two-moment calculation, special root-length test or isolated signed-case file is needed. All signed certificates use the positive-indicator normalization, rather than retaining separate ordinary and adjoint-parity enumeration modes in the proof allocation.

9.2 Proof of Theorem 1.1

The pinned repository census [5] gives the counts in Table 1. Section 8 supplies the seventeen unitary realizations and their exact tensor matches. For every other ring with $\mu \le 17$, the exhaustive ledger supplies a general exclusion certificate covered by the theorems and cited inputs above. There are 23396 such certificates. At multiplicity eighteen, all 5080 rings have general exclusion certificates; the last two are the composite inert-prime parity/spherical certificates of Corollary 7.2. At exact multiplicities nineteen and twenty, every one of the $5640 + 6866$ repository tensors has a general exclusion certificate. These later certificates use the same criterion order and exact checker, but are addressed directly against the pinned repository tensor order, not the older local basis labels. The resulting 40982 exact general exclusions, together with the seventeen unitary realizations, prove the theorem. □

The exhaustive step here is a finite, exact certificate verification; it is not a proof-assistant formalization of the categorical theorems. The release includes every tensor, branch and integer witness, so its finite assertions can be reproduced from the pinned census rather than trusted as a numerical census label. The two proof-ledger verification commands recheck the entire $28493 + 12506 = 40999$-ring source and each accepted certificate.

μ	To decide	D	Cy	Sub	C4	QP	QS	IP	Gate	Sign	Open
1	6	6	0	0	0	0	0	0	0	0	0
2	34	26	0	0	0	0	0	0	4	4	0
3	80	62	0	2	0	2	0	0	6	8	0
4	132	119	0	0	0	0	0	0	2	11	0
5	209	197	2	1	0	0	0	0	0	9	0
6	336	312	0	1	0	0	0	0	4	19	0
7	477	461	0	1	0	0	0	0	0	15	0
8	733	708	0	0	1	0	0	0	1	23	0
9	863	848	1	0	0	0	0	0	1	13	0
10	1082	1064	0	0	0	1	0	0	2	15	0
11	1383	1365	0	1	0	0	0	0	2	15	0
12	1948	1920	0	2	1	1	1	0	1	22	0
13	2211	2188	0	0	0	2	3	0	0	18	0
14	2554	2531	1	1	0	0	1	0	3	17	0
15	2936	2909	1	2	0	1	1	0	0	22	0
16	4157	4125	0	1	1	0	0	0	1	29	0
17	4255	4234	0	0	0	5	0	0	0	16	0
18	5080	5049	1	0	0	0	0	2	2	26	0
19	5640	5609	1	2	0	3	0	0	0	25	0
20	6866	6825	1	0	1	5	0	0	1	33	0

Table 4: Exact-multiplicity exclusion sequence. To decide omits independently constructed unitary models. Stage symbols follow the row order in Table 3; each ring appears once, at its first accepted certificate. Open concerns general categorification by these criteria.

9.3 The two exceptional tensors

In the twenty self-dual coordinates of Section 2, the two last tensors requiring a new criterion are

$$\begin{aligned} v^{(5073)} &= (16,16,17,17,17,17,17,15,17,17,\\ &\qquad 17,17,17,16,17,18,18,17,17,18),\\ v^{(5076)} &= (17,16,16,17,16,17,17,17,17,17,\\ &\qquad 17,16,17,17,17,17,17,17,18,18). \end{aligned} \tag{26}$$

They have common sorted FP type

$$(1,a,a,a+1,a+2), \qquad a=33+2\sqrt{285}, \qquad D=9120+540\sqrt{285}.$$

Their formal-codegree polynomials are respectively

$$\begin{aligned} &(t-3)^2(t-15)(t^2-18240t+68400),\\ &(t-3)(t-5)^2(t^2-18240t+68400). \end{aligned}$$

Both reduce to $t^2(t+1)^3$ modulo two. Consequently Lemma 5.1 restricts a hypothetical sphericalization's entire dimension field to $\mathbb{Q}(\sqrt{285})$. Since $285 \equiv 5 \pmod 8$ and the irrational coefficient is two, Theorem 7.1 applies, as checked in Corollary 7.2. The general exclusions are the composite exact certificates, independent of any signed-table timeout.

9.4 Numbering and reproducibility

The release's notation $F^{(5)}_{m,j}$ preserves the first index and basis from the historical ancillary file of [10] when one exists. Classes absent from that file have appended archival labels in

μ	Total	Unitary models	General exclusions	General residuals
1	16	10	6	0
2	37	3	34	0
3	82	2	80	0
4	134	2	132	0
5	209	0	209	0
6	336	0	336	0
7	477	0	477	0
8	733	0	733	0
9	863	0	863	0
10	1082	0	1082	0
11	1383	0	1383	0
12	1948	0	1948	0
13	2211	0	2211	0
14	2554	0	2554	0
15	2936	0	2936	0
16	4157	0	4157	0
17	4255	0	4255	0
18	5080	0	5080	0
19	5640	0	5640	0
20	6866	0	6866	0

Table 5: Exact automatic coverage of all pinned rank-five tensors through multiplicity twenty. All general, pivotal, spherical, pseudo-unitary and unitary existence questions in this range are decided.

local tensor order. Separately, the field `local_index` gives the one-based ordinal within the archived tensor records at fixed m. At multiplicity eighteen the two indices coincide. Fingerprints of the full tensors, rather than these labels, key every certificate.

The command `python verify.py` checks the first 28493 tensors, canonical uniqueness, the full pinned-repository isomorphism crosswalk, and all 28476 general exclusion certificates, the seventeen constructed model matches and both inert-prime parity/spherical composite certificates. The companion command `python software/verify_extension.py` checks exact source order, ring axioms, uniqueness and all 12506 later general exclusion certificates. Both regenerate each finite matrix and signed branch. Python/native Gram comparisons and deliberate certificate mutations are additional controls. The command `python -O verify.py` performs the same proof checks: none is implemented using a removable Python `assert`. The accepted separating vectors are tested with integers; a numerical optimization solver is used only in discovery.

The verification guide gives exact commands, input conventions, checked integer ranges, dependencies and the failure policy. The bundled pinned repository tensors are the census input. The discovery command accepts either an external tensor or a canonical index and applies the same criteria at the chosen moment height and search budget. The archive also contains complete decision and provenance ledgers, input hashes and provenance notes.

AI use disclosure

The research, code, and manuscript were generated with AI assistance. Independent human verification and Lean formalization are pending.

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